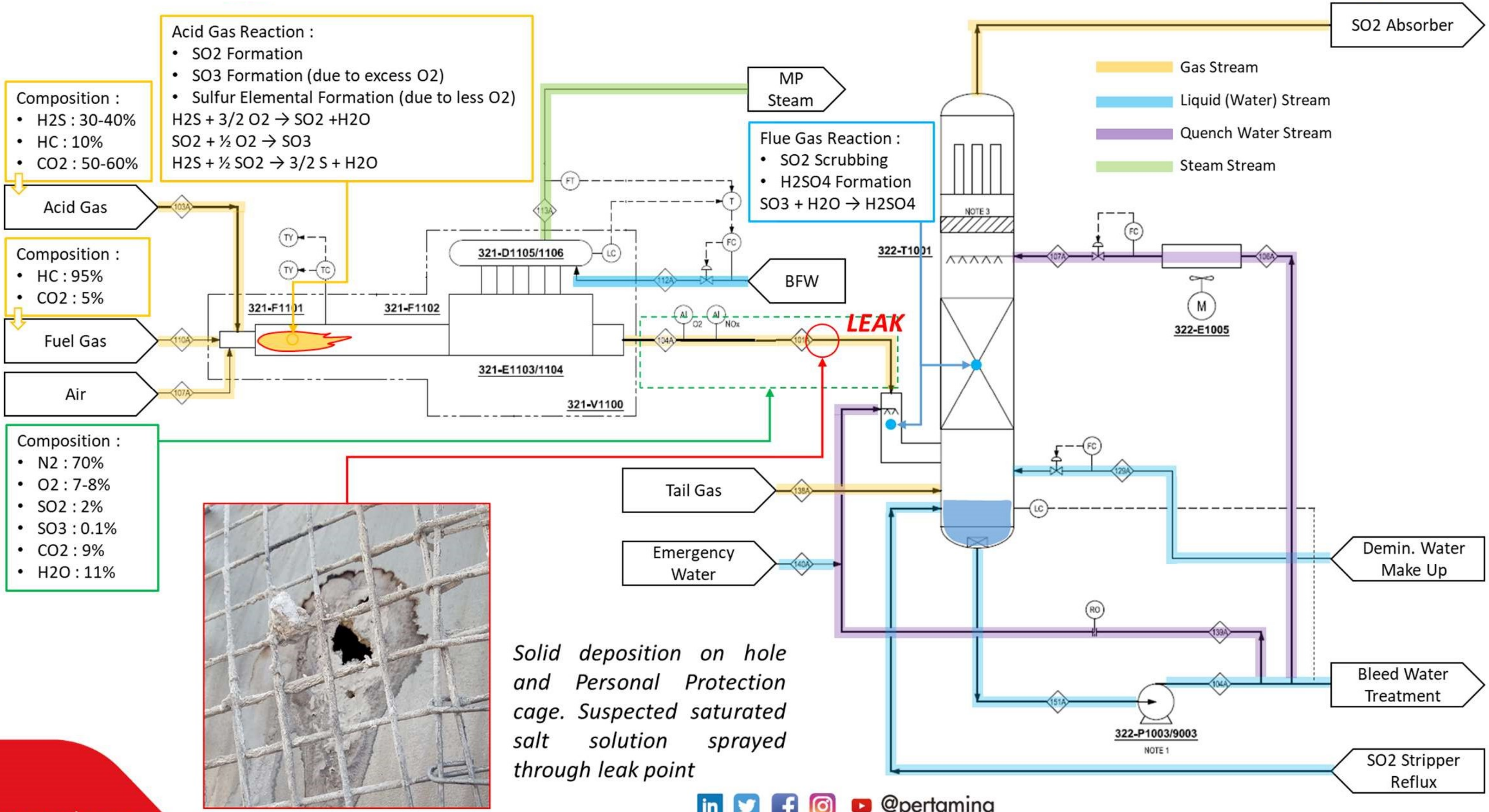
Investigation & RCA for 321 AGI, 48” Flue Gas Line Leak.

Preliminary Finding Analysis

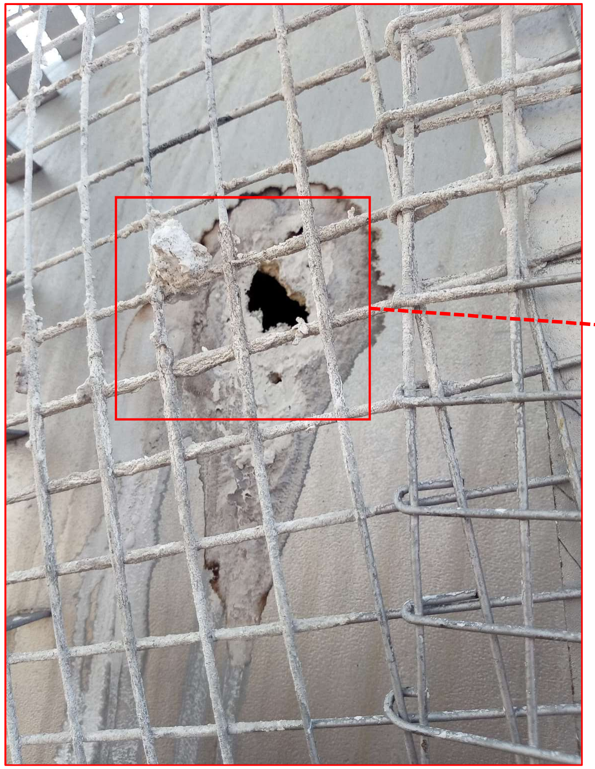
19 September 2024

321 – Acid Gas Incinerator Train A, 48" Flue Gas Line Leak Overview



Solid deposition on hole and Personal Protection cage. Suspected saturated salt solution sprayed through leak point

Investigation : AGI Flue Gas line Train A



Solid Analysis

2. Soluble in Water

Quantitative Analysis :			
Total Iron, Fe	%-w	APHA 3111B	26.91
Sulfate, SO4	%-w	ASTM D4327-03	50.75
Zinc, Zn	%-w	APHA 3111B	0.0030

3. Insoluble in Water

Qualitative Analysis :			
Sulfur Test	Negative		
Quantitative Analysis :			
Total Iron, Fe	%-w	APHA 3111B	0.0963
Aluminium, Al	%-w	APHA 3111B	0.0004
Zinc, Zn	%-w	APHA 3111B	<0.0001

4. Physical Properties

Moisture	%-w	ASTM D2216	22.24

Comment *The solubility results indicate that 99.68% of the substance was dissolved in demineralized water
*The analysis of moisture is conducted using the original sample

High solubility in water indicating the solid was high purity salt, FeSO4

Analysis :

- Proven by solid lab analysis result, its confirmed that high possibility of damage mechanism 48" flue gas line was corrosion due to existence of H2SO4 inside flowing stream.
- Solid deposition appears on leak area and Personal Protection cage, suspected deposition mechanism occur through spraying salt saturated solution from leak point.
- Spraying fluid from leak point indicating system on pressurized condition.

Comparison Facts : AGI Flue Gas line Train B

AGI & AGE Train B has offline since Mid 2023 due to Candle Filter High DP for Sulfur Elemental deposition inside of Candle Filters.

Mid 2023

During AGI Train B reinstatement August 2024, its found that Vent line on 48” Flue Gas line has been disintegrated, suspected due to corrosion.

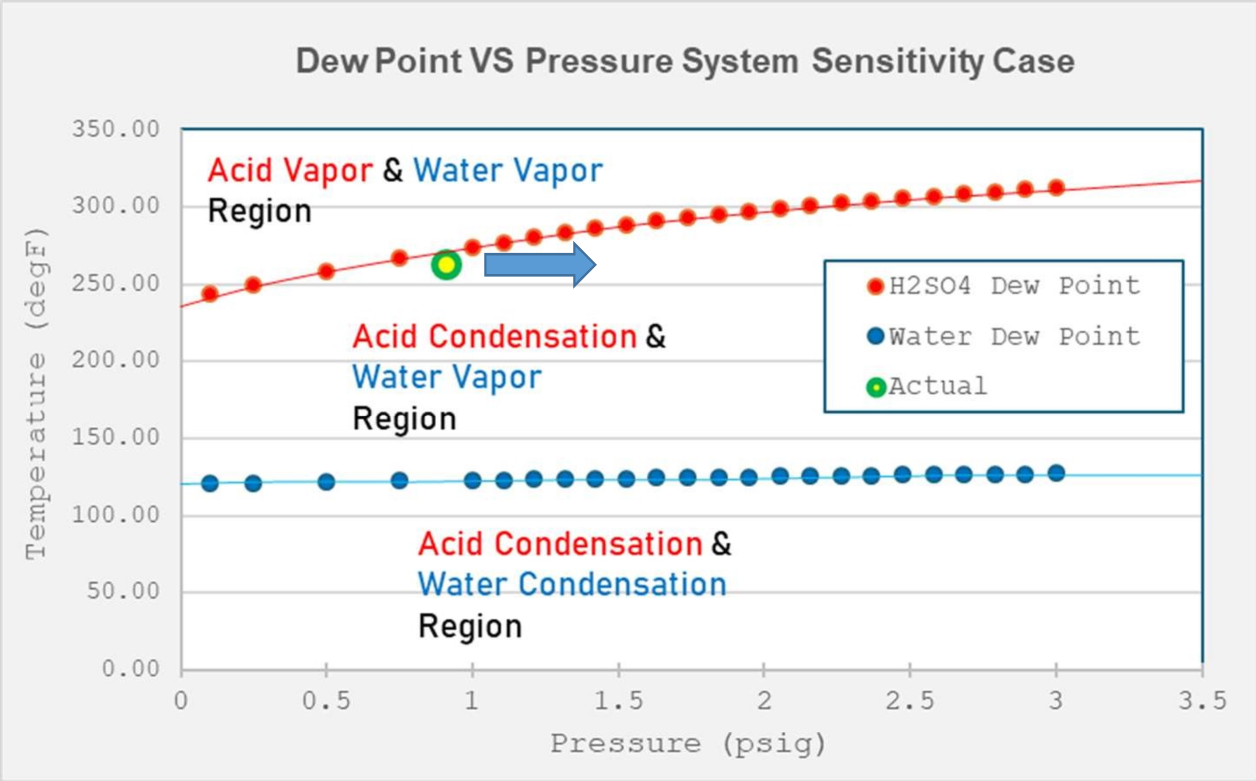
The solid analysis prove that highly possible vent line damaged due to corrosion of H2SO4 exist on flue gas line.

August 2024

2. Ash Analysis			
Insoluble Material in HCl	%-w	Gravimetric	0.9583
Soluble Material in HCl			As Elemental
Iron, Fe	%-w	APHA 3111B	31.5037
Chromium, Cr	%-w	APHA 3111B	<0.0001
Magnesium, Mg	%-w	APHA 3111B	0.0071
Copper, Cu	%-w	APHA 3111B	0.0550
Zinc, Zn	%-w	APHA 3111B	0.0099
Lead, Pb	%-w	APHA 3111B	<0.0001
Potassium, K	%-w	APHA 3111B	0.0279
Sodium, Na	%-w	APHA 3111B	0.0355
Calcium, Ca	%-w	APHA 3111B	0.1335
3. Physical Properties			
Moisture	%-w	ASTM D2216	13.4552
Loss on Ignition @ 750 °C	%-w	ASTM D7348	63.3878
Ash Content	%-w	ASTM D7348	36.6122
4. Chemical Analysis			
Sulfate, SO4	%-w	ASTM D4327-03	61.2562

Operability Analysis : AGI Flue Gas line Train A, Dew Point Analysis

Dew Point for H2SO4 pure & H2O Pure for stream dew point analysis boundary.



Operating Temperature and Pressure analysis on leak area :

- Acid Dew Point curve was using pure H2SO4, which conservative for upper boundaries while pure water act as lower boundaries. <u>Actual Acid Dew Point curve shall be within upper and lower boundaries</u>.
- Operating condition during incident : 0.91 psig (from PZT inlet Pre-Scrubber A) & 270 degF as average calculated temperature from skin temperature indication near leak area. Actual pressure might be higher if no leakage occur on upstream PZT.
- Suspected current operating condition on saturation condition.
- Acid Condensation occur significantly, suspected due to Increase System Pressure.
- As Pressure System increase, Acid Dew Point temperature will increase accordingly, means vapor service will easier to condense.
- Higher pressure on flue gas line will occur as impact from Block Filter case in 322 Pre-Scrubber.
- Hypothesis : Failure on vent tapping point 48" flue gas line AGI train B, due to corrosion, as impact from higher system pressure, as impact from Candle Filter blockage.

Operability Analysis : AGI Flue Gas line Train A, Candle Filter Performance

Candle Filter Performance Evaluation

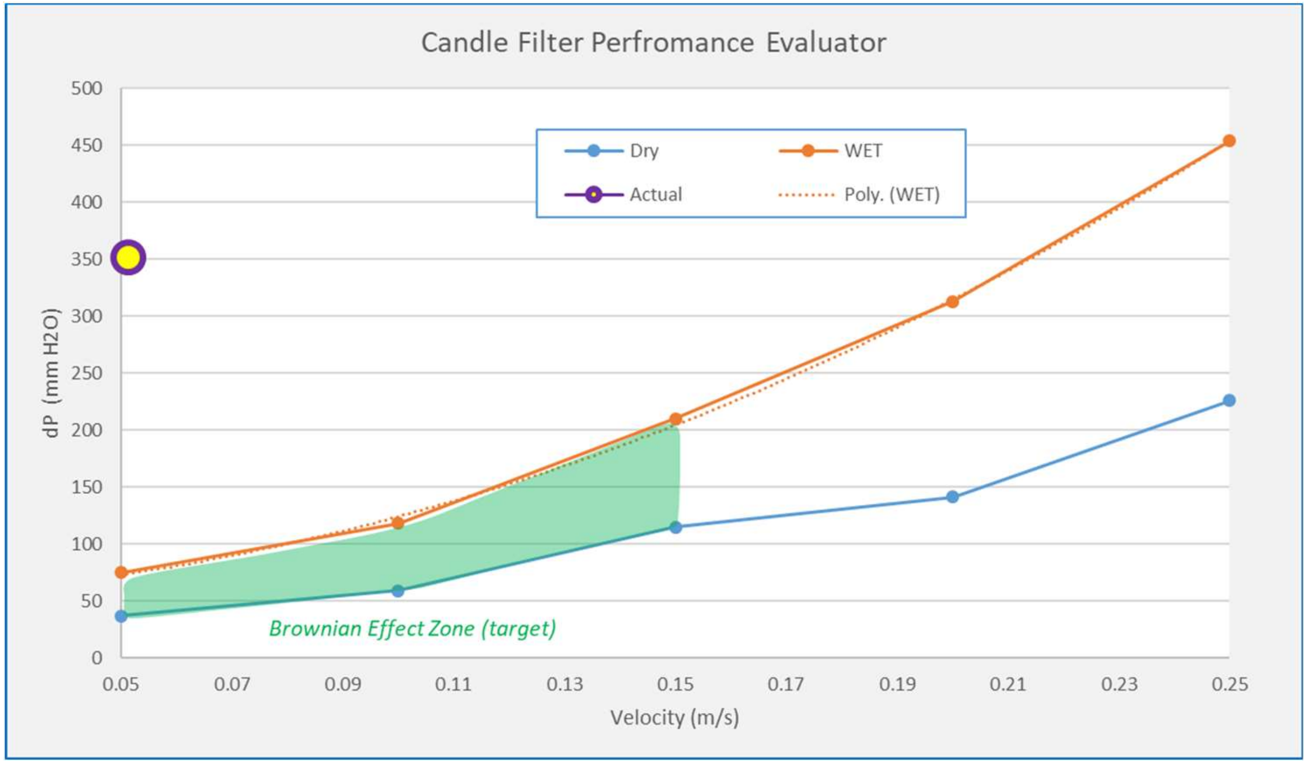
Velocity		m/s	0.05149046	m/s
phi			3.142857143	
Radius		m	0.2335	m
Area Inlet		m2	-	m2
Edge Inlet		m	1.467714286	m
Area Filter		m2	7.338571429	m2
Single Element Candle Filter				
Flow		m3/s	0.377866417	m3/s
		m3/hr	1360.3191	m3/hr
		ft3/hr	48039.21569	ft3/hr
		ft3/day	1152941.176	ft3/day
Candle Filter Package (17 ea)				
Flow		ft3/day	19600000	ft3/day
		MMSCFD	19.60	MMSCFD
Acid Gas Incenerator Ratio (1 : 13), MAXIMUM				
Air Flow		MMSCFD	18.20	MMSCFD
Acid Flow		MMSCFD	1.40	MMSCFD
Total		MMSCFD	19.60	MMSCFD

Goal Seek

Goal Seek

Velocity	0.051490	m/s
P brownian	74.0621995	mm H2O
Active Area	21.068%	

Relative Load	34.326%
Blockage	**61.376%**



- Candle Filter Pre-Scrubber A indicating lower performance due to reducing active area for mist handling.
- Active area reduction impacted by blockage. Calculated blockage ratio already reaching 61% (assumed candle filter performed as per manufacture test reference).
- Blockage on candle filter will give back pressure up to upstream process.

Root Cause Analysis - Cause & Effect Method : 48" AGI Flue Gas Line Leak

Leak on 48" AGI Flue Gas line → Corrosion and/or Erosion → Acidic Liquid exist on flue gas stream → Condensation of flue gas on pipe → Higher Pressure in 48" Flue Gas pipe → Blockage on Candle Filter

Leak on 48" AGI Flue Gas line

- SO2 release
- Its appears that saturated salt liquid sprayed from hole

Corrosion and/or Erosion

- Solid deposition on area of hole exist (salt like)
- Confirmed has high solubility on water (99% water soluble)
- Fe 27%, SO4 51% w/w.
- Suspected iron sulfate salt appears on hydrous salt form.
- Hydrous salt form exist due to water molecule trapped inside crystal structure

Acidic Liquid exist on flue gas stream

- Lead by finding of Sulfate ion (=SO4) on solid sample analysis
- Highly possible acidic corrosion of carbon steel due to H2SO4 occur.
- H2SO4 exist due to side reaction of acid gas (H2S) combustion thru advanced SO2 oxidation then resulting SO3 formation then reacted with condensed water (from combustion reaction)
- $H_2S + O_2 = SO_2 + H_2O$
- $SO_2 + O_2 = SO_3$
- $SO_3 + H_2O = H_2SO_4$

Condensation of flue gas on pipe

- 35% H2S Concentration from Acid Gas feed stream will be combusted and resulting same amount of H2O as side product.
- Amount of H2O exist in acid gas combustion flue gas could increased in presence of fuel gas.
- H2O as result of combustion shall stay at vapor condition at designed flue gas temperature (around 500 degF) and 1.4 psig.

Higher Pressure in 48" Flue Gas pipe

- Temperature element indicated 450 degF. At this temperature, won't give significant condensation.
- Suspected higher operating pressure, as result of back pressure from Pre-Scrubber (lower temperature due to quenching process & candle filter performance).
- Based on HYSYS simulation, Pressure has significant impact to dew point of H2SO4 and simulated flue gas.

Blockage on Candle Filter

- Reduction of Candle Filter performance due to blockage indication, 0.5-0.6 psi of dP (higher dP than designed, at 0.35 psi for high alarm).
- Calculated blockage or inactive area of candle filter at 61% from design performance curve.
- High dP of Candle filter resulting Higher Back Pressure to upstream process (AGI section), increasing dew point temp of upstream fluid subsequently increasing tendency of Condensation.

Improving Facts : Historical evidence from Process 2 Tr. B, 48" flue gas line vent plug corrosion event. Even with lower running time, indicating worst corrosion rate. Plugged candle filters on pre-scrubber B was observed before. It's convincing for Back pressure Hypothesis in regards with acid dew point corrosion damage mechanism.